

Dr Morgan John Brand

Generalist Scientist · Microbiology & Systems — imagine < create < change

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Summary

A generalist scientist and PhD graduate. I own and live on a stretch of the KwaZulu-Natal coast; I grow bananas on it my own way, and I'm less interested in farming the land than in coming to understand it — reading it closely enough to work with how it already behaves. Microbiology and systems are things I experience here first-hand, not only study. Trained through aquaculture and microbiology, I think in systems: soil, water, data, and the microbial life that binds them together. I move easily between the lab, the field, and the codebase, with work spanning reproducible data science, applied AI, geospatial analysis, and the living systems of the land itself. What connects it is a scientist's instinct for how a whole system behaves, and a builder's habit of turning messy real-world data into something clear enough to act on.

Experience

Owner & Grower — Canaan · Marina Beach, KwaZulu-Natal · 2016–present

My own land on the KwaZulu-Natal coast — I live on it and grow bananas by my own methods.

- Grow, harvest, pack, and move bananas
- Irrigation systems and growing-field design
- Financial management and record keeping

MSc / PhD Researcher — University of Cape Town · Cape Town · 2011–2019

Postgraduate research training (aquaculture, microbiology, data science).

- Awarded NRF Research Scholarship for scarce skills
- Outreach educator for schools
- Part-time lecturing and teaching of undergraduate coursework

Education

Doctor of Philosophy (PhD) — Science

University of Cape Town · 2023

Thesis: Ulva as a functional feed: a practical investigation into the effects of Ulva on the growth, consumption, health, and microbiome of the farmed abalone *Haliotis midae*.

BSc Honours — Ichthyology & Fisheries Science

Rhodes University · 2010

Thesis: Evaluation of algae, grown in brewery effluent water, as a potential dietary ingredient for the farmed abalone *Haliotis midae*.

BSc — Zoology & Ichthyology

Rhodes University · 2009

Skills

- **Data Science in R** — tidyverse, data pipelines, reproducible analysis, ggplot2
- **Reproducible Research** — Quarto, Git / GitHub, renv, Nix, literate programming
- **Applied AI** — LLM-assisted research & analysis, data & literature synthesis, scientific writing support
- **Data Engineering** — pipeline design, data capture & cleaning, tabular & structured data
- **Geospatial Analysis & GIS** — GDAL / GeoPandas / Rasterio, MapLibre & PMTiles, DEM, stream & watershed analysis, cartography
- **Microbiology** — microbiome analysis
- **Aquaculture** — abalone, seaweed / Ulva, system design
- **Land & living systems** — growing bananas, irrigation & water, soil & land systems, contextual land observation
- **Research** — experimental design, scientific writing

Selected Publication

Bolton JJ, Cyrus MD, Brand MJ, Joubert M, Macey B (2016). **Why grow Ulva? Its potential role in the future of aquaculture.** *Perspectives in Phycology*. 10.1127/pip/2016/0058